

WORK COMPLETED BY

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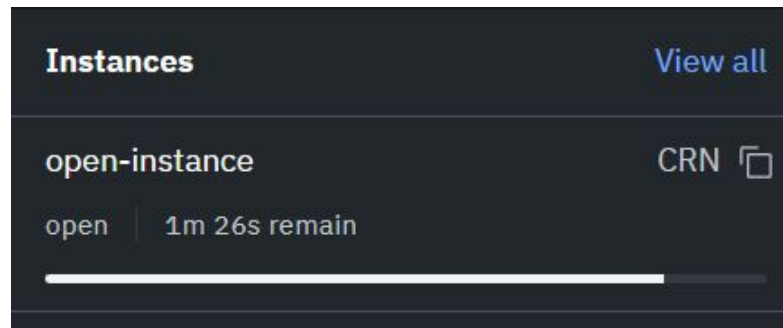
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IBM QPU TIME ESTIMATION

Objective: To determine a formula to estimate time taken by the QPU to simulate a molecule.

Need for time estimation

- On IBM Quantum Platform the free plan only provides 10 minutes of Quantum Runtime.
- These 10 minutes very quickly becomes the limiting factor when working with larger and more complex molecules
- In order to solve this issue we can request IBM to provide some Quantum Runtime for research purposes.
- For this reason we are to estimate the QPU time by creating a formula.



Runtime Usage on IBM Quantum Platform

Progress made so far

$$N_c \cdot S (d \cdot t_g + t_r + t_d + t_i)$$

Where,

N_c = No. of grouped circuits,

S = No of shots

d = circuit depth

t_g = avg gate time, t_d = repetition delay

t_r = avg readout time, t_i = initialization time

Results

A few more runs were then done by switching out multiple parameters such as,

- Number of shots
- The ansatz (which controlled the circuit depth)
- The Active space (which controlled the circuit depth and no of grouped circuits)

A common pattern we observed were this formulas is good for over-estimation of quantum time when concerned with linear molecules. Although when concerned with cyclical molecules this pattern breaks.

Next Steps

For now, rather than just getting data from a pure quantum scenario, we have decided to shift focus to comparison between Noiseless and Noisy Classical simulations with the Quantum simulation.

The Noiseless runs are just to be there as a baseline but the true comparison will happen between Noisy and Quantum simulation.

The Noisy simulation can be brought closer to match the Quantum simulation by changing the Noise model object used to define the Estimator.

Noise Model

- **Single-Qubit Errors:** Introduces a 0.1% depolarizing error and applies it to standard 1-qubit gates (like h, rx, u1).
- **Two-Qubit Errors:** Introduces a larger 1% depolarizing error and applies it specifically to the 2-qubit cx (CNOT) gate.
- **Readout Errors:** Simulates a measurement error where a qubit has a 98.5% chance of being measured correctly, and a 1.5% chance of flipping (measuring a 0 as a 1, or vice versa)
- **Estimator Setup:** Packages the entire noise model into an AerEstimator configured to run 20,000 shots.

```
from qiskit_aer.primitives import Estimator as AerEstimator
from qiskit_aer.noise import NoiseModel, depolarizing_error,
ReadoutError

noise_model = NoiseModel()

error_1q = depolarizing_error(0.001, 1)
error_2q = depolarizing_error(0.01, 2)
noise_model.add_all_qubit_quantum_error(error_1q, ['rx', 'ry', 'rz',
'h', 'u1', 'u2', 'u3'])
noise_model.add_all_qubit_quantum_error(error_2q, ['cx'])

probabilities = [[0.985, 0.015], [0.015, 0.985]]
readout_error = ReadoutError(probabilities)
noise_model.add_all_qubit_readout_error(readout_error)

noisy_estimator = AerEstimator(
    backend_options={"noise_model": noise_model},
    run_options={"shots": 20000}
)
```

THANKYOU